HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed quantitative trading models in **Python**, integrating **Pandas** and **NumPy** for data manipulation and feature engineering on sports datasets.
- Executed trades on prediction markets using an in-house model, focusing on maximizing expectancy and meeting defined business objectives.
- Refined trading strategy logic for execution and market-making, directly contributing to improved trading performance and identifying pricing inefficiencies.
- Performed post-trade analysis to evaluate model performance and market behavior, providing actionable insights for strategy development and calibration.
- Collaborated with quantitative researchers and engineers to integrate model enhancements and deploy trading infrastructure improvements, enhancing overall system efficiency.
- Took product ownership of specific trading strategies, defining priorities, implementing features, and optimizing end-to-end workflows for the trading desk.
- Containerized trading tools and data pipelines using **Docker** and deployed them on **AWS EC2** instances for scalable and reliable execution.

SKILLS

Programming Languages – Python, SQL, R

Data Analysis & Libraries – Pandas, NumPy, Scikit-learn, Matplotlib

Trading & Markets – Prediction Markets, Quantitative Trading, Market Making, Algorithmic Trading, Strategy Development

Cloud & DevOps – AWS (EC2, S3, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile, Scrum, REST APIs

PROJECTS

Quantitative Football Trading Strategy

- Developed a Python-based quantitative model to analyze player performance and tactical data for football matches, aiming to identify betting market inefficiencies.
- Engineered data pipelines using **Pandas** and **SQL** to ingest and process historical match statistics, player ratings, and betting odds from multiple sources.
- Backtested the trading strategy against historical data, achieving a simulated return of approximately **18%** over a two-year period with a Sharpe ratio of **1.2**.

Sports Prediction Market Analysis Tool

- Built a data visualization tool in **Python** using **Matplotlib** and **Seaborn** to analyze market dynamics and identify potential arbitrage opportunities in sports prediction markets.
- Integrated real-time market data feeds and historical performance metrics to provide actionable insights for trading decisions.
- Implemented automated data collection scripts that processed over **50,000** data points per week, reducing manual analysis time by **40%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)